

Felipe Eduardo Farias Belo

Senior Software Engineer

Passionate about **backend, database systems, and embedded systems**, I thrive on exploring new technologies and embracing ongoing challenges. Throughout my tenure at Virtus (CEEI/UFCG), I worked on research and development projects demanding continuous adaptation. I proactively absorbed and shared knowledge, applying best practices like **Scrum, Clean Architecture**, and **Test-Driven Development**. I also mentored junior members and undergraduate students, supporting their growth and commitment to quality. I aim to deliver scalable, robust solutions aligned with evolving tech trends and software engineering excellence.

Professional Experience

Virtus Research Center for Innovation in Information Technology, Communication and Automation - CEEI/UFCG 2015 - 2025

Moura — Senior Software Engineer — Backend
Team Technical Lead

2025

- Defined the initial backend architecture for a MES solution.
- Set up Docker containers for development, testing, and production environments.
- Focused on scalability, clarity, and maintainability.

Skills: TypeScript, NestJS, PostgreSQL, Git, Docker/Containers, Scrum, Software Architecture

Ericsson — Senior Software Engineer — Technical
Lead (Unity VR)

2024 – 2024

- Led the Unity VR app for robot control using Meta Quest 3.
- Implemented compute shaders for point-cloud processing and rendering inside the headset.
- Designed immersive VR UI/UX and defined the system architecture.

Skills: Unity, C#, Compute Shaders, VR UX, Git, System Architecture, Point-Cloud Rendering

Positivo — Senior Software Engineer — Backend Developer

2023 – 2024

- Built REST APIs and PostgreSQL schemas for a React + .NET Framework web app.
- Ensured client-server integration and wrote unit tests for reliability.

Skills: .NET Framework, C#, PostgreSQL, REST APIs, React Integration, Unit Testing, Git

Epson — Senior Software Engineer — Backend Team Technical Lead

2020 – 2023

- Developed a proof-of-concept backend using Java Spring Boot and PostgreSQL.
- Integrated with three different front-end applications and deployed on Raspberry Pi.
- Created REST APIs for multiple clients and optimized for constrained hardware.

Skills: Java, Spring Boot, PostgreSQL, REST APIs, Embedded Systems, Raspberry Pi, System Architecture

Inovatic — Senior Software Engineer — Front-end Team Technical Lead

2019 - 2020

- Led a front-end team for network equipment control (muxponders, transponders).
- Managed MIBs (SNMP) and YANG data models; integrated Grafana dashboards.
- Oversaw a new Angular + Flask application and deployment on embedded Linux systems.
- Maintained and extended a production legacy system in Django (front-end & back-end).

Skills: Angular, Flask, Django, Linux Server, Embedded Systems, Grafana, MIBs (SNMP), YANG Models, Leadership

TukLuk — Middle Software Engineer — Full Stack Developer

2018 – 2019

- Worked on a Management Information System (MIS) for R&D (PID) projects with tax incentives.
- Developed features using Java, Spring Boot, Angular, and PostgreSQL.
- Integrated Google Drive API for user file downloads and contributed to system architecture.

Skills: Java, Spring Boot, Angular, PostgreSQL, Google Drive API, Full Stack, System Architecture

VALID — Middle Software Engineer — Android Developer

2017 – 2018

- Built a white-label Android client (UI) for a prototype mobile solution.
- Developed intermediate UI versions for feature testing and verification flows.
- Designed the app architecture to support future customer-specific UIs.

Skills: Android, Java, White-label Mobile, Secure UI Design, Git, Prototyping

Asus — Middle Software Engineer — Backend Developer

2017 – 2017

- Delivered backend features for a global production project.
- Profiled database queries and optimized performance.

Skills: Python, Django, PostgreSQL, Git, Scrum, Software Architecture

Asus — Middle Software Engineer — Android Developer

2017 - 2017

- Developed an Android application deployed globally within the company.

Skills: Java, Android, REST API, Software Architecture, Git, Scrum

Compal Electronics — Middle Software Engineer — Unity/Game Developer (Technical Lead)

2016 – 2017

- Led Unity development of a network-based game for Windows Surface Table and PCs.
- Built core animations and features; implemented OCR for piece recognition.

Skills: C#, Unity, OCR, Software Architecture, Git, Scrum

Asus — Junior Software Engineer — Game Developer

2015 – 2016

- Contributed to an Android game to promote Asus devices.
- Created an AAR package to secure sensitive code components.
- Refactored code for best practices and implemented new features.
- Collaborated on the architecture for a parallel game project.

Skills: Java, C#, Unity, Android, Git, Architecture, Sensitive Data Security, Scrum

Embedded—Laboratory of Embedded Systems and Pervasive Computing 2013 – 2015

University — Undergraduate Student

- Trained by Sony in Android technologies and company devices, participating in hands-on projects during the program.
- Participated in the development of a smart home system with sensors controlled via an Android app and a web front-end; the sensors were managed by embedded systems.
- Participated in the development of a mobile OCR application for text extraction from images, leveraging Android ecosystem libraries.
- Participated in the development of a Sony smartwatch app to support tennis training, capturing athlete movement data.

Skills: Android, Java, Database, OCR, Web Front-End, Smartwatch Apps, Embedded Systems, Scrum, Git

BELADIA—Research Laboratory in BioEnergy, Dedicated Architectures, and Artificial Intelligence 2014 – 2015

University — Undergraduate Student

- Participated in the development of a C application using fuzzy logic to assess thermoelectric plant emissions against applicable legal thresholds.
- Designed linguistic variables and fuzzy rules to classify/indicate compliance and support operational decision-making.
- Integrated the algorithm with continuous emissions monitoring (CEMS) data for near real-time analysis.

Skills: C, Fuzzy Logic, CEMS/Continuous Emissions Monitoring, Signal Processing, Environmental Compliance, Git, Scrum

Education

- Electrical Engineering (incomplete)—UFCG (2003–2008)
- Bachelor of Computer Science (incomplete) - UFPB (2008-2012)
- Digital Games Technology Degree - Facisa (2010-2013)

Skills

Languages & Frameworks

- TypeScript, JavaScript
- Java, Spring Boot
- C#, .NET Framework
- Python, Django, Flask
- C, C++
- Angular
- Unity (VR/AR)
- NestJS

Backend & Databases

- PostgreSQL
- REST APIs
- Database Systems
- Microservices Architecture
- System Integration

DevOps & Tools

- Git, CI/CD
- Docker/Containers
- Linux Server
- Raspberry Pi

Specialties

- Embedded Systems
- Android Development
- VR/AR Development
- OCR & Image Processing
- Compute Shaders
- Fuzzy Logic
- SNMP/MIBs, YANG Models
- Grafana, Monitoring

Practices & Methodologies

- Software Architecture
- Clean Architecture
- Scrum, Agile Methodologies
- Technical Leadership
- Full Stack Development

Contact

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